

CHAPTER 2

PRINCIPAL COMPONENT ANALYSIS

Throughout this book, we use $\underline{x}$ to denote a vector. In case the vector is a p -dimensional vector, we have that

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}.$$

Vectors are always column vectors and we denote the space of all p -dimensional vectors by $\mathbb{R}^p$. We also use the symbol T to denote the transpose of a vector.

$$\underline{x} = (x_1, x_2, \dots, x_p)^T.$$

We use the bold notation to denote a matrix. For example, the $n \times p$ matrix with entry $x_{i,j}$ on row i and column j is denoted by $\mathbf{X}$ and:

$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,p} \\ x_{2,1} & x_{2,2} & \dots & x_{2,p} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & \dots & x_{n,p} \end{pmatrix}.$$

If $\mathbf{A}$ is an $n \times p$ matrix, then we denote by $\mathbf{A}\underline{x}$ (or $\mathbf{A} \times \underline{x}$) the usual matrix multiplication:

$$\mathbf{A}\underline{x} = \begin{pmatrix} \sum_{j=1}^p a_{1,j}x_j \\ \sum_{j=1}^p a_{2,j}x_j \\ \vdots \\ \sum_{j=1}^p a_{n,j}x_j \end{pmatrix}.$$

The dot-product of two p -vectors is defined as follows:

$$\underline{x} \cdot \underline{y} = \sum_{i=1}^p x_i \times y_i.$$

The norm of a vector quantifies the length of the vector and we denote the norm by $\|\underline{x}\|$. The norm can be determined as follows:

$$\|\underline{x}\|^2 = \underline{x} \cdot \underline{x}.$$

If we want to determine the distance between two observations, we use:

$$\text{distance between } \underline{x} \text{ and } \underline{y} = \|\underline{y} - \underline{x}\|. \quad (2.1)$$

In Chapter 9, we will give a more general overview of vectors, dot-products and norms, but for now this suffices to introduce the theory of principal component analysis.

2.1 Example 1: a first illustration

Consider a two dimensional space with two observations, A and B . The coordinates of A and B are:

$$A = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

In order to identify these two observations, we need to know both coordinates. That is, if in the future we want to locate the points A and B , we need to store for each of the two observations an x -coordinate and a y -coordinate. The two observations are shown in Figure 2.1 by the two dots. This is of course a simplified situation. In reality, one will have

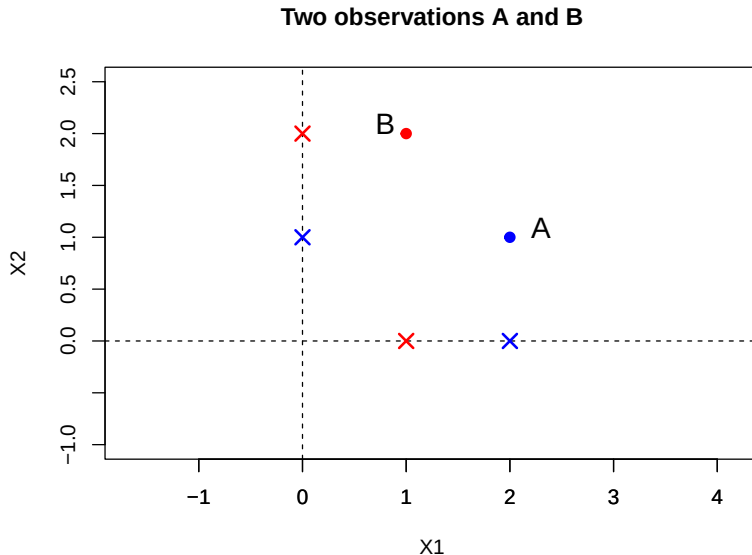


Figure 2.1: The two observations $A(2, 1)$ and $B(1, 2)$ used in Section 2.1 are represented by the blue and the red dot, respectively. The blue crosses indicate the orthogonal projections of A on the x and y axis. The red crosses are the orthogonal projections of B on the x and y axis.

more than two observations and to identify a particular observation, we need more than two coordinates. For example, assume we have a car insurer who has a data set with information about the different policyholders. Each policyholder has a set of characteristics, such as the age, location, mileage, etc. Storing all the characteristics of the different policyholders will require space. Moreover, finding information about a particular policyholder will take time, especially if the data set is large.

To understand the idea of *principal component analysis*, we have to understand the concept of an *orthogonal projection*. Assume we have a point A and a line g . The point $\underline{z}$ is the orthogonal projection of A on the line g if $\underline{z}$ is the point on g for which the distance to A is minimized.

For example, we can take the line g as either the x or the y axis. It is then straightforward to see that the orthogonal projection of the points A and B on the x and y axis are as follows

- A on the x axis: $(2, 0)^T$.
- B on the x axis: $(1, 0)^T$.
- A on the y axis: $(0, 1)^T$.
- B on the y axis: $(0, 2)^T$.

Indeed, if we, for example, determine the orthogonal projection of A on the x axis, then we search for a point $\underline{z}$ on the x axis, such that the line through $\underline{z}$ and A is orthogonal with the x axis. The orthogonal projections of A and B are indicated in Figure 2.1 with red crosses (for B) and blue crosses (for A).

We define two new vectors, $\underline{\alpha}_1$ and $\underline{\alpha}_2$. Note that for the moment we do not consider the question of how these vectors are determined. However, we will provide a detailed discussion on why we use exactly these vectors. The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are given by

$$\underline{\alpha}_1 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^T \quad \text{and} \quad \underline{\alpha}_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^T.$$

One can now check the following properties.

- The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are normalized:

$$\|\underline{\alpha}_1\| = \|\underline{\alpha}_2\| = 1.$$

- The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are orthogonal:

$$\underline{\alpha}_1 \cdot \underline{\alpha}_2 = 0.$$

In Figure 2.2, we show the two observations A and B in blue together with $\underline{\alpha}_1$ and $\underline{\alpha}_2$ in red. We also draw a red dashed line through the special vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$. The dashed blue lines are the ‘normal’ (or the

usual) axes. The two red lines are orthogonal as well as the two blue lines. Moreover, if we take the two dashed blue axes, we can rotate them until they coincide with the dashed red lines. Otherwise stated, the two dashed red lines are essential the same as the dashed blue lines, up to a rotation.

The dashed red line which goes through $\underline{\alpha}_1$ is denoted by f_1 and given by

$$f_1(x) = -x.$$

If we denote the orthogonal projection of A on the line f_1 by A' , one finds that

$$A' = \left(\frac{1}{2}, -\frac{1}{2} \right)^T. \quad (2.2)$$

Similarly, the orthogonal projection B' of the point B on this line is given by

$$B' = \left(-\frac{1}{2}, \frac{1}{2} \right)^T.$$

We observe that the orthogonal projections A' and B' of A and B are different.

We can repeat the exercise, but now use a different line. The line f_2 given by

$$f_2(x) = x$$

corresponds with the line running along the vector $\underline{\alpha}_2$. The orthogonal projections of A and B on f_2 are the same. Indeed, if we have a look at Figure 2.2, we can clearly see that if we start in A and we go orthogonal until we cross the line f_2 , we end up in the same point as when we do the same exercise, but now starting in B . As a result, unlike the orthogonal projections of A and B on f_1 , the orthogonal projections of the two different points A and B on f_2 result in only one observation.

Assume that we rotate the whole figure 2.2 counterclockwise until the f_2 line points upward. We can then do as if the f_2 line serves as the y axis and the f_1 line as the x axis. However, note that the new x axis now points to the left and not to the right, since its direction is determined

by the direction of the vector $\underline{\alpha}_1$. We hence created a new coordinate system. We call the new coordinate system Z_1 and Z_2 . What needs to be done is to express the points A and B in the new coordinate system. Indeed, the x coordinate of the point A is 2 and the y coordinate is 1, but these numbers 2 and 1 correspond to the coordinates needed to identify the point A in the ‘old’ coordinate system, i.e. when the dashed blue lines serve as the two axes to locate the points.

The first coordinate of A in the new coordinate system is related to the norm of the orthogonal projection of A on Z_1 . The second coordinate is related to the norm of the orthogonal projection on Z_2 . In fact, to locate the point A using the red coordinate system, we need to know how far we have to follow each of the new (red) axes. Therefore, the sign of the coordinates in the new coordinate system depends on the direction of the orthogonal projections. For example, the orthogonal projection of A on f_1 is on the ‘negative’ side. In fact, the vector A' , determined in (2.2), points in the opposite direction as the vector $\underline{\alpha}_1$. Therefore, the Z_1 coordinate of A will be negative. We can then determine the Z_1 coordinate of the point A by determining the distance from $(0, 0)$ to A' :

$$Z_1 = - [\text{Distance from } (0, 0) \text{ to } A'],$$

with the help of Expression (2.1), we find

$$Z_1 = -\sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = -\frac{1}{\sqrt{2}}. \quad (2.3)$$

However, it is interesting to notice that there is an alternative calculation using the inner product and $\underline{\alpha}_1$:

$$Z_1 = \underline{\alpha}_1 \cdot A = -\frac{1}{\sqrt{2}}. \quad (2.4)$$

Note that this method is much easier to use than using the distance from $(0, 0)$ to A' . In a later moment, we will show that it is not a coincidence that formula (2.4) gives exactly the same results as formula (2.3). We conclude that the new coordinate Z_1 of the point A can easily be determined using the ‘basis vector’ $\underline{\alpha}_1$.

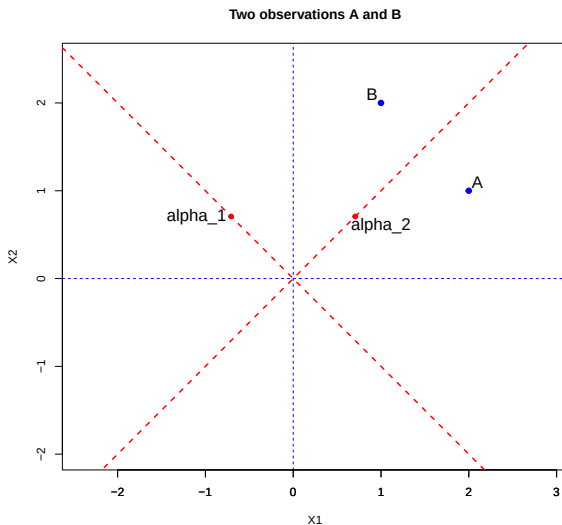


Figure 2.2: The two observations A and B together with the special vectors α_1 and α_2 .

One can check that the points A and B in the new coordinate system are

$$A = \left(-\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right)^T \text{ and } B = \left(\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right)^T.$$

We see that the second coordinate of A and B in this new system is the same. Otherwise stated, the points A and B are different in the Z_1 direction, but the same in the Z_2 direction.

Assume we want to ‘travel’ from A to B , where we are only allowed to travel along the axes, meaning we can only move parallel to the two axes. In the old coordinate system, we need to move parallel to the x and the y axis. Moreover, we need two steps. In the new coordinate system we can move along the red axes. But now it is sufficient to only move along the Z_1 axis. In this sense, the red system is easier, or less complex, than the blue system. This is the most important takeaway from this section: changing our coordinate system can make the points in our data set look less complex. Note also that we did not change the

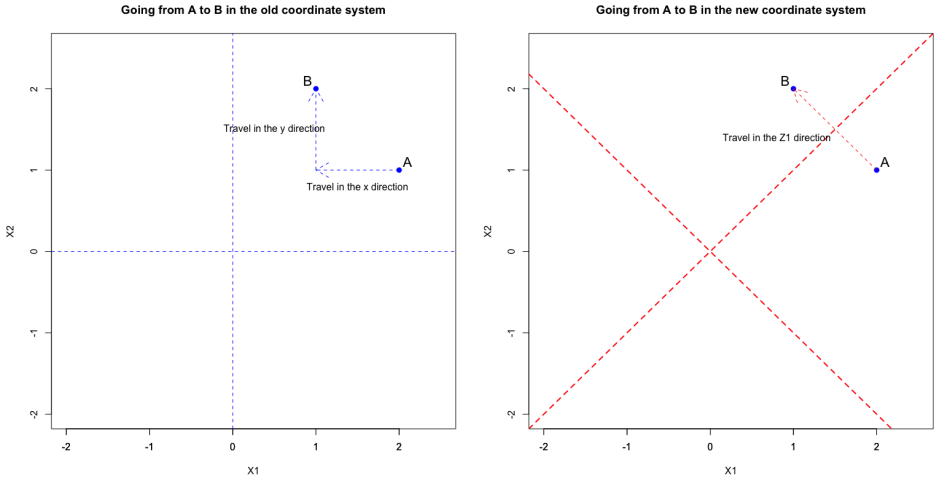


Figure 2.3: Traveling from points A to B in two different coordinate systems.

data, i.e. the observations A and B , we only changed the coordinate system. If we want to be able to distinguish the two observations, we only need to store the Z_1 coordinates of A and B , whereas in the ‘old’ coordinate system, we need to store both. Hence, we can reduce the storage space by 50% when we use the red system to identify our data points A and B .

In this chapter about principal component analysis, we show how to determine the vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$, such that we can rotate the coordinate system and reduce the complexity of a data set.

2.2 Example 2: dependent random variables

We will use a second example to confirm our findings of Example 1 in a more realistic setting. Assume we have observations from a two dimen-

sional random vector:

$$(X_1, X_2)^T = \text{bivariate normal.}$$

This means that the random variables X_1 and X_2 are assumed to be standard normal:

$$X_i = N(0, 1), \text{ for } i = 1, 2.$$

The random variables X_1 and X_2 are dependent:

$$\text{Cov}[X_1, X_2] = \sigma_{1,2}.$$

If the covariance $\sigma_{1,2}$ is positive, the two random variables are more likely to be moving together, whereas they tend to move in opposite directions if the covariance $\sigma_{1,2}$ is negative. Note also that in this setting where the random variables X_1 and X_2 are normal, the random variables are independent if, and only if, the covariance is zero. The R code [2.2.1](#) generates two dependent random variables which both have a normal distribution with mean 0, variance 1 and a covariance equal to 0.8. The code works as follows. We start with generating a first random variable X_1 having a normal distribution with mean 0 and variance 1. In order to have a second random variable X_2 which is dependent on X_1 , we take a linear combination between X_1 and an independent random variable, say Z as follows:

$$X_2 = \rho X_1 + \sqrt{1 - \rho^2} Z.$$

The covariance is then exactly equal to ρ . Note that since we work with random variables with variance 1, the correlation coincides with the covariance. If we would like to work with non-standardized normal distributions, we can scale the random variables X_1 and X_2 in such a way they have the desired mean and standard deviation.

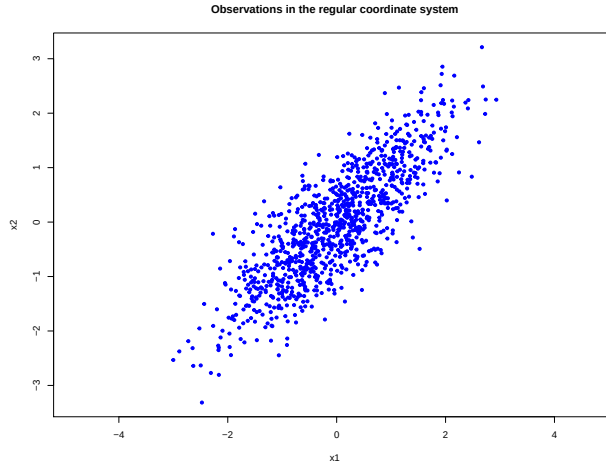


Figure 2.4: A scatterplot with 1000 realizations of the random vector $(X_1, X_2)^T$ which are generated by the R code 2.2.1.

R Code 2.2.1: Generating a data set

```
set.seed(20)
# simulate the first variable
w=rnorm(1000,0,1)
# create the second variable
v=0.8*w+sqrt(1-0.8^2)*rnorm(1000)
# store the data in a data frame
X=data.frame(x1=w, x2=v)
```

Assume for the moment the situation where one faces a situation where one can work with *independent* normal random variables. If one wants to simulate from a set of independent random variables, each normal random variable can be simulated independently from the other ones. Hence, in the bivariate case, one could proceed with the following steps to generate independent random variables:

- Step 1: Determine realizations for X_1 from a normal distribution.
- Step 2: Determine realizations for X_2 from a normal distribution.

Note that one could even switch step 1 and step 2, and start with generating realizations from X_2 and then determine the realizations for X_1 . If one needs to add a third independent normal random variable, we have to add a step 3, where we generate realizations from this X_3 random variable. Therefore, the complexity of the algorithm increases linear with the number of random variables. Independence of random variables makes life much easier. In this particular case, we showed that the complexity of Monte Carlo simulation scales linear with the number of variables you have, as long as they are independent.

The situation is different when we consider *dependent* random variables. Assume we want to simulate a new observation from $(X_1, X_2)^T$ where now the two random variables are dependent. The steps we need to follow are now:

- Step 1: Determine realizations for X_1 from a normal distribution.
- Step 2: Determine realizations for X_2 such that:
 - the realizations for X_2 come from a normal distribution,
 - the dependence structure between X_1 and X_2 is maintained.

Note that in this case, we cannot reverse step 1 and step 2, because step 2 uses the results of step 1. The algorithm to determine realizations from dependent random variables is more complex than the independent case. We now need to simultaneously determine the two random variables. In particular, step 2 of the algorithm is more complex now compared to the independent case. Indeed, we additionally have to ensure the dependence structure is maintained between X_2 and X_1 . Moreover, assume we add a third random variable, which is dependent on both X_1 and X_2 . In this case, we need to add a third step which is even more complex than the second step of the algorithm. Indeed, we have to determine realizations for X_3 such that the dependence structure with both X_1 and X_2 is maintained.

Assume we, similar to the example in Section 2.1, define two vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$:

$$\underline{\alpha}_1 = (0.7048, 0.7094)^T \text{ and } \underline{\alpha}_2 = (0.7094, -0.7048)^T.$$

The choice of the two vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ looks at the moment arbitrary. We will in the following sections explain how to determine these vectors. For now, we want to illustrate that these vectors will allow us to simplify this particular data set.

In Figure 2.5, we have plotted the observations of the normal random variables X_1 and X_2 . We can draw the line which runs through $\underline{\alpha}_1$ and the line which runs through $\underline{\alpha}_2$. These lines are in Figure 2.5 shown in red. The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are, similar to the situation in Example 2.1, orthogonal. Therefore, we can again move from the original axes (in blue) to the red axes (in red) by rotating the axes. The observations can then be expressed in terms of the new, red, axes using the following formulas:

$$Z_1 = 0.7048 \times X_1 + 0.7048 \times X_2, \quad (2.5)$$

$$Z_2 = 0.7094 \times X_1 - 0.7048 \times X_2. \quad (2.6)$$

In R box 2.2.2, we show the observations of the data set in the original coordinate system, i.e. using the X_1 and X_2 values. For example, the first observation has coordinates $(-0.0145, 0.0365)$. We also determine the covariance matrix, which shows that both X_1 and X_2 have the same variance (equal to 1) and a correlation which is approximately 0.8. We also express the data set using the Z_1 and Z_2 values. The interesting part here is that determining for each observation a Z_1 and Z_2 value is easy once we know the special vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$. Indeed, we only need to take a linear combination of the X_1 and X_2 values of each observation. The first observation of our data set expressed in the new coordinate system is $(0.0157, -0.0360)$. We also determine the covariance matrix. What we see is that, unlike the X_1 and X_2 variables, Z_1 has a larger variance than Z_2 . Secondly, we observe that the Z_1 and Z_2 variables are uncorrelated.

Remember that we discussed above that working with independent random variables has several (mathematical) advantages. However, the vector (X_1, X_2) is, based on the data, not independent. Using the ‘trick’ with the rotation, we transfer the data into a new data set, such that (Z_1, Z_2) is now independent. Therefore, the data set expressed using the Z_1 and Z_2 values is easier and less complex than the same data, but expressed using X_1 and X_2 .

The total variation, or variability, of the data set is defined as the variance in all directions. However, which choice of coordinate system we use, should not affect the total variation of the data. Indeed, by switching the axes, we look in a different way at the same data. Therefore, we have that

$$\text{Var}[X_1] + \text{Var}[X_2] = \text{Var}[Z_1] + \text{Var}[Z_2].$$

In this case, we also know that $\text{Var}[X_1] = \text{Var}[X_2]$, hence the data has the same variability in both directions. However, we also have that $\text{Var}[Z_1]$ is much larger than $\text{Var}[Z_2]$. The total variation of the data set is mainly explained by Z_1 , and to a much lesser extend by Z_2 .

Variability of a data set is important, since it allows us to make a meaningful prediction model. Indeed, if we observe that the variables X_1 and X_2 have a large variance, it means that we have heterogeneous data set. If we want to use these variables to predict some outcome, say Y , then we can hope that different outcomes for Y are caused by the heterogeneity of the data. We employ a statistical or machine learning method to detect the relationship. On the other hand, assume you have a data set with a low variance. In this case, the different observations look all similar, and will most likely not help to explain the different observations of a certain target variable Y .

Assume we look at the data using the X_1 and X_2 representation. Moreover, assume that, for some reason, we can only store one variable. We have to sacrifice 50% of the data. We prefer to keep most of the variability in the new, reduced data set, since it will allow us to build a

prediction model. In this case where we use the X_1 and X_2 variables to identify the data, there is no clear way to distinguish X_1 and X_2 in terms of variability. Indeed, they have equal variance. Otherwise stated, the choice does not matter since we will always lose 50% of the total variability. If we use the Z_1 and Z_2 values of the data set, the picture looks different. The obvious choice is to keep Z_1 and remove Z_2 . Indeed, *although we remove 50% of the data (all Z_2 values), we keep $1.785/2 = 89\%$ of the total variability.*

R Code 2.2.2: The data set in the new coordinate system

```

# The observations in the normal coordinate system
head(X)
      x1      x2
1 -0.01451361  0.03658752
2  0.36978945  1.00939147
3  0.31180349  0.36331659
4 -0.21309816 -0.66970941
5 -0.01940631  0.60224750
6 -0.85919877 -1.24983272

var(X)
      x1      x2
x1 0.9897767 0.7953137
x2 0.7953137 0.9910908

# The special vectors
alpha1=c(0.7048, 0.7094)
alpha2=c(0.7094, -0.7048)

# Determine the Z1 and Z2 transformation
Z1=X$x1*alpha1[1]+X$x2*alpha1[2]
Z2=X$x1*alpha2[1]+X$x2*alpha2[2]

# The observations in the new coordinate system
head(data.frame(Z1=Z1, Z2=Z2))
      Z1      Z2
1  0.0157260 -0.03608284
2  0.9766899 -0.44909047
3  0.4774959 -0.03487214
4 -0.6252834  0.32083936
5  0.4135568 -0.43823087
6 -1.4921946  0.27136649

# determine the variance and covariances
cov(data.frame(Z1=Z1, Z2=Z2))
      Z1      Z2
Z1 1.78571948 0.00451675
Z2 0.00451675 0.19513094

```

the random variables Z_1 and Z_2 are both normal distributed, with

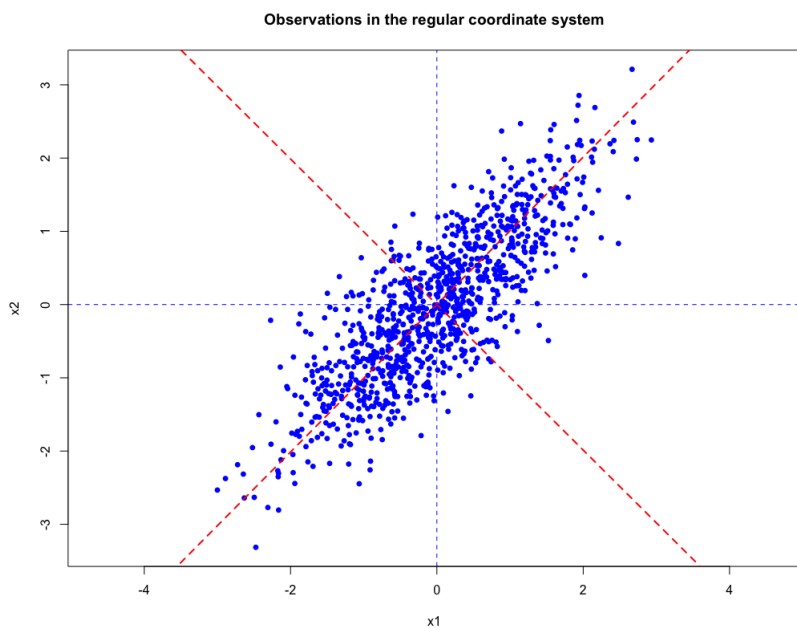


Figure 2.5: The observations of the random variables X_1 and X_2 together with the original axes (in blue) and the new axes (in red).

mean zero, but with different variances. Since the correlation is zero, it is straightforward to simulate new realizations for $(Z_1, Z_2)^T$. Using (2.5), we then also find

$$(X_1, X_2)^T = A^{-1} \times (Z_1, Z_2)^T,$$

where

$$A = \begin{pmatrix} \alpha_{1,1} & \alpha_{2,1} \\ \alpha_{1,2} & \alpha_{2,2} \end{pmatrix}.$$

This implies that we can determine values for $(X_1, X_2)^T$ by first simulating from the independent vector $(Z_1, Z_2)^T$, and then take a linear combination using the inverse matrix A^{-1} .

The matrix A is called the rotation matrix. One can use A to determine the values of the data set in the new coordinate:

$$Z = AX.$$

We use A^{-1} to recover the original values of the data set, when we have the new values. Moreover, one can prove that for the situation where the matrix A contains the special vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$, the inverse A^{-1} is equal to the transpose A' . As a result, we can recover the original data from the new data as follows

$$X = A^T X.$$

We will later discuss how to determine the vector $\underline{\alpha}_1$, but we can already provide an interpretation. Indeed, remember that

$$\underline{\alpha}_1 = (-0.7048, -0.7094).$$

Using Expression (2.5), we find that if the variables X_1 and X_2 both increase/decrease together, the variable Z_1 decreases/increases. As a result, the variable Z_1 captures the **co-movements** of the original data set X_1 and X_2 . The vector $\underline{\alpha}_2$: as given by

$$\underline{\alpha}_2 = (0.7094, -0.7048)^T.$$

If X_1 increases and X_2 decreases, the variable Z_2 increases. As a result, the variable Z_2 captures the **counter-movements** of X_1 and X_2 .

We can conclude that changing the coordinate system, allows us to transform the original data (X) to a new data set (Z). This new data set is less complex, has most of the variance shifted on the first variable Z_1 and the new variables have a clear interpretation.

2.3 PCA: finding the new coordinate system

In the previous examples, we showed that one can switch from the old coordinate system to a new coordinate system by doing a linear transformation. Moreover, we do not change the data during the process, but we only change the perspective and look at the data from a different angle. This change of perspective simplifies the data and helps us to identify which variables are more important and which variables, if needed, can be omitted. In this section, we will show how one arrives at this special coordinate system that is used to transform the data.

We work with a data set with p different dependent variables. Therefore, consider the p -dimensional random vector:

$$\underline{X} = (X_1, X_2, \dots, X_p)^T.$$

We assume the data is standardized, i.e.:

- the mean is zero: $\mathbb{E}[X_i] = 0$,
- and the variance is one: $\text{Var}[X_i] = 1$.

Note that you can always transform the data such that it is standardized. The different components of the random vector are dependent:

$$\text{Cov}[X_i, X_j] = \sigma_{i,j}.$$

We define the Variance-Covariance matrix Σ as follows

$$\Sigma = \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \dots & \sigma_{1,p} \\ \sigma_{2,1} & \sigma_{2,2} & \dots & \sigma_{2,p} \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{p,1} & \sigma_{p,2} & \dots & \sigma_{p,p} \end{pmatrix}.$$

The random variables X_i and X_j are assumed to have variance 1, and therefore the covariance $\sigma_{i,j}$ is equal to the correlation between the two random variables.

Similar as in the two previous examples, we want to define a new coordinate system, such that the data is not expressed using the variables $(X_1, X_2, \dots, X_p)^T$, but instead we use the new system $(Z_1, Z_2, \dots, Z_p)^T$. Switching between the two sets of variables corresponds with a rotation of the axes. Moreover, we want that each Z_i can be determined by taking an appropriate *linear* combination of the variables $X_1, X_2, \dots, X_p$. Our approach will be to start with investigating how to determine the values for Z_1 , then we move to Z_2 , etc.

The first new component Z_1 of our data set has to be a linear combination of the ‘old’ variables, hence we already know that there exist $\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,p} \in \mathbb{R}$ such that

$$Z_1 = \alpha_{1,1}X_1 + \alpha_{1,2}X_2 + \dots + \alpha_{1,p}X_p. \quad (2.7)$$

The question is now:

How can we determine $\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,p}$?

In the original data set, we have p variables, but each variance is equal (to one). This means that variations in the, say, X_1 direction are of equal magnitude (on average) than in any other direction. However, going back to Figure 2.4 in Example 2.2, shows us that it would be interesting to transform the data set, such that most of the variability happens in the Z_1 direction. Otherwise stated:

We take the $\alpha_{1,j}$ such that the *variance of Z_1 is as large as possible*.

The variance of Z_1 is defined in (2.7) and can be expressed as follows:

$$\text{Var}[Z_1] = \text{Var}[\alpha_{1,1}X_1 + \alpha_{1,2}X_2 + \dots + \alpha_{1,p}X_p]. \quad (2.8)$$

In vector notation, we can write

$$\text{Var}[Z_1] = \underline{\alpha}_1^T \Sigma \underline{\alpha}_1,$$

where we have that $\underline{\alpha}_1 = (\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,p})^T$, and Σ is the variance-covariance matrix.

The variance in (2.8) is an increasing function in $\alpha_{1,j}$. This means that we can always increase the variance of Z_1 , by increasing one of the weights. Therefore we impose the following condition on the vector weights

$$\alpha_{1,1}^2 + \alpha_{1,2}^2 + \dots + \alpha_{1,p}^2 = 1.$$

If we use the vector $\underline{\alpha}_1$ to gather all the weights we can write this condition in vector notation:

$$\underline{\alpha}_1 \cdot \underline{\alpha}_1 = 1.$$

We will search for the best possible weight vector $\underline{\alpha}_1$ with unit length to maximize the variance of Z_1 . We can formulate this problem as follows:

The maximization problem to find $\underline{\alpha}_1$.

$$\max_{\underline{\alpha}_1} \underline{\alpha}_1^T \Sigma \underline{\alpha}_1 \quad \text{subject to} \quad \underline{\alpha}_1 \cdot \underline{\alpha}_1 = 1. \quad (2.9)$$

In the theorem below, we provide the solution to our problem

Theorem 2.3.1 *The solution $\underline{\alpha}_1$ of the maximization problem (2.9) satisfies the following relation:*

$$\Sigma \underline{\alpha}_1 = \lambda \underline{\alpha}_1. \quad (2.10)$$

Proof. We solve the maximization problem by using the Lagrange function, which we denote here by f :

$$f(\underline{\alpha}_1) = \underline{\alpha}_1^T \Sigma \underline{\alpha}_1 - \lambda (\underline{\alpha}_1 \cdot \underline{\alpha}_1 - 1).$$

We optimize the function f by setting the first derivative with respect to $\underline{\alpha}_1$ equal to zero. We have that

$$\frac{\partial f}{\partial \underline{\alpha}_1} = 2(\Sigma \underline{\alpha}_1 - \lambda \underline{\alpha}_1).$$

Therefore, we find that

$$\frac{\partial f}{\partial \underline{\alpha}_1} = 0,$$

leads to the relation (2.10). Moreover, we can verify that this indeed leads to a maximum by checking the second order conditions. ■

The result in Theorem 2.3.1 can be interpreted in the context of eigenvectors and eigenvalues, which are fundamental concepts in linear algebra. Eigenvalues often quantify the scaling effect of a transformation, while eigenvectors indicate the directions in which this scaling occurs. Indeed, we have the following important definition in linear algebra.

Definition 2.3.1 *Let A be an $n \times n$ matrix. A scalar λ is called an **eigenvalue** of A , and a nonzero vector $\underline{v} \in \mathbb{R}^n$ is called an **eigenvector** of A corresponding to λ , if they satisfy the equation*

$$A\underline{v} = \lambda\underline{v}.$$

Since the covariance matrix Σ and the vector $\underline{\alpha}_1$ are satisfying (2.10), we find that $\underline{\alpha}_1$ is an eigenvector of Σ and the corresponding λ is the eigenvalue of the eigenvector $\underline{\alpha}_1$.

Note, however, that Theorem 2.3.1 does not provide an answer to the question of how to determine the vector $\underline{\alpha}_1$. Indeed, we find that the vector $\underline{\alpha}_1$ has to satisfy condition (2.10), but this does not yield a unique solution. For a general $p \times p$ covariance matrix Σ , there are several eigenvectors and eigenvalues, so several choices for $\underline{\alpha}_1$ that all satisfy (2.10). The question thus reduces to: which one should we take?

Recall that if we find an eigenvector $\underline{\alpha}_1$ and eigenvalue λ_1 of the covariance matrix Σ , then the variance of Z_1 can be expressed as follows

$$\text{Var}[Z_1] = \underline{\alpha}_1^T \Sigma \underline{\alpha}_1 = \lambda_1 \underline{\alpha}_1^T \underline{\alpha}_1 = \lambda_1,$$

where the second equality follows because $\underline{\alpha}_1$ satisfies (2.10) and the third equality since we required that the vector $\underline{\alpha}_1$ has norm 1. Therefore, if we have multiple eigenvectors and eigenvalues, we have to take the eigenvector with the *largest* eigenvalue.

In order to calculate the second principal component, we search for a new variable Z_2 which is, similar to Z_1 , a linear combination of the existing variables:

$$Z_2 = \alpha_{2,1}X_1 + \alpha_{2,2}X_2 + \dots + \alpha_{2,p}X_p.$$

We want that the variance of Z_2 is again as large as possible. However, we already know that the variance of Z_2 cannot be larger than the variance of Z_1 . For the same reason as in the calculation of the first principal component, we have the technical condition that the norm of $\underline{\alpha}_2$ should be equal to 1. However, for this second principal component, we add an important condition, which states that Z_1 and Z_2 should have covariance equal to zero. This is inspired by Figure 2.5, where we showed that by rotating the axes, we end up with two new directions such that the data is uncorrelated in the new system.

The random variable Z_2 is defined as a linear combination of the original data:

$$Z_2 = \alpha_{2,1}X_1 + \alpha_{2,2}X_2 + \dots + \alpha_{2,p}X_p.$$

We have that the vector $\underline{\alpha}_2$ should have length 1, i.e.

$$\underline{\alpha}_2 \cdot \underline{\alpha}_2 = 1.$$

We also require that the random variables Z_1 and Z_2 are uncorrelated, which means that

$$\text{Cov}[Z_1, Z_2] = 0.$$

We can show that the covariance can be expressed using the vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ as follows

$$\text{Cov}[Z_1, Z_2] = \underline{\alpha}_2^T \Sigma \underline{\alpha}_1 = \lambda_1 \underline{\alpha}_2^T \underline{\alpha}_1.$$

We can formulate the problem of finding the second principal component as follows:

The maximization problem to find $\underline{\alpha}_2$.

$$\max_{\underline{\alpha}_2} \underline{\alpha}_2^T \Sigma \underline{\alpha}_2 \text{ subject to } \underline{\alpha}_2 \cdot \underline{\alpha}_2 = 1 \text{ and } \underline{\alpha}_1 \cdot \underline{\alpha}_2 = 0. \quad (2.11)$$

Theorem 2.3.2 *The solution $\underline{\alpha}_2$ of the maximization problem (2.11) satisfies the following relation:*

$$\Sigma \underline{\alpha}_2 = \lambda \underline{\alpha}_2. \quad (2.12)$$

Proof. The Lagrange function to find the $\underline{\alpha}_2$ is given by

$$f(\underline{\alpha}_2) = \underline{\alpha}_2^T \Sigma \underline{\alpha}_2 - \lambda (\underline{\alpha}_2 \cdot \underline{\alpha}_2 - 1) - \phi \underline{\alpha}_2 \cdot \underline{\alpha}_1.$$

In order to find the optimum, we determine the derivative:

$$\frac{\partial f}{\partial \underline{\alpha}_2} = 2\Sigma \underline{\alpha}_2 - 2\lambda \underline{\alpha}_2 - \phi \underline{\alpha}_1.$$

Putting the derivative equal to zero and multiplying both sides with the vector $\underline{\alpha}_1$ we found in Theorem (2.3.1) gives

$$2\underline{\alpha}_1^T \Sigma \underline{\alpha}_2 - 2\lambda \underline{\alpha}_1^T \underline{\alpha}_2 - \phi \underline{\alpha}_1^T \underline{\alpha}_1 = 0. \quad (2.13)$$

Since $\underline{\alpha}_1$ corresponds with an eigenvector of the covariance matrix Σ , we have that $\underline{\alpha}_1^T \Sigma = \lambda_1 \underline{\alpha}_1^T$. Therefore, the first term of expression (2.13) can be rewritten as follows:

$$2\underline{\alpha}_1^T \Sigma \underline{\alpha}_2 = 2\lambda_1 \underline{\alpha}_1^T \underline{\alpha}_2.$$

If we require that the solution $\underline{\alpha}_2$ is orthogonal to the first principal component $\underline{\alpha}_1$, we have that $\underline{\alpha}_1^T \underline{\alpha}_2 = 0$ and therefore the first and second term in (2.13) will be zero. Moreover, we required that the solution $\underline{\alpha}_1$ is normalized, so we find from (2.13) that

$$\phi = 0.$$

The optimum $\underline{\alpha}_2$ of this function is thus determined by solving the following equation:

$$\Sigma \underline{\alpha}_2 = \lambda \underline{\alpha}_2.$$

■

We conclude that once we have determined the eigenvectors and eigenvalues of the covariance matrix, the second principal component corresponds with the eigenvector that belongs to the second largest eigenvalue. The other principal components can be determined in similar ways. Note, however, that deriving the solution becomes more complex, since there are more covariance constraints.

Theorem 2.3.3 *The k -th principal component is the eigenvector $\underline{\alpha}_k$ corresponding with the k -th largest eigenvalue λ_k . Moreover, we have that*

$$\text{Var}[Z_k] = \lambda_k, \text{ for } k = 1, 2, \dots, p. \quad (2.14)$$

Recall that the original data set consists of p variables $X_1, X_2, \dots, X_p$. We additionally assumed that each variable X_i has variance 1. The total variability of the data set is then defined as the aggregate variability in the different directions. Otherwise stated, the total variability of the original data set is

$$\text{Total Variability} = \sum_{i=1}^p \text{Var}[X_i] = p. \quad (2.15)$$

We have developed a methodology to change the original coordinate system in which we used the variables $X_1, X_2, \dots, X_p$ to a new coordinate system in which we use the variables $Z_1, Z_2, \dots, Z_p$. Note, however, that by switching from the original coordinate system the new coordinate system in which we use the principal components, we are actually expresses the **same** data in a **different** way. The total variability is thus the same after the rotation:

$$\sum_{i=1}^p \text{Var}[X_i] = \sum_{i=1}^p \text{Var}[Z_i]. \quad (2.16)$$

On the level of the variability of the total data set, nothing really happened when we apply a rotation and switch to a new coordinate system.

However, expressing the data with the new variables $Z_1, Z_2, \dots, Z_p$ has advantages. Firstly, the total variability is evenly divided over the different variables when using the original variables $X_1, X_2, \dots, X_p$. This same aggregate variability is unevenly divided over the new variables $Z_1, Z_2, \dots, Z_p$. Indeed, we have that

$$\text{Var}[Z_1] > \text{Var}[Z_2] > \dots > \text{Var}[Z_p]. \quad (2.17)$$

Taking into account (2.14), the proportion of the total variance explained by the k -th principal component can then be defined as follows:

$$\text{Variance explained by } Z_k = \frac{\lambda_k}{\sum_{i=1}^p \text{Var}[X_i]}. \quad (2.18)$$

Assume, for example that we only use the first m principal components:

$$\text{Variance explained by } m \text{ components} = \frac{\sum_{k=1}^m \lambda_k}{\sum_{i=1}^p \text{Var}[X_i]}. \quad (2.19)$$

If we use the variables $Z_1, Z_2, \dots, Z_p$, we need all p to ‘explain’ the total variability. However, using the variables $Z_1, Z_2, \dots, Z_p$ instead of $X_1, X_2, \dots, X_p$ has the advantage that we are guided when we need to take a subset of the variables. Indeed, assume we use the variables $X_1, X_2, \dots, X_p$, but we need to reduce the data set and we can only keep one variable. Which of the p variables should we take? If we have to reduce the data, then we aim to keep as much variability as possible. Since each variable X_i has variance 1, there is no best choice when selecting one variable in case we use the variables $X_1, X_2, \dots, X_p$.

If we use the variables $Z_1, Z_2, \dots, Z_p$, however, the story is very different. Indeed, in this case the choice is obvious. Since Z_1 has the largest variance, we best keep the variable Z_1 . Similarly, using only m principal components results in a ‘loss’ of information. but if the first m principal components explain a ‘sufficient large proportion’ of the total variance, reducing the number of dimensions from p to m may be a good approximation of the data.

2.4 Application: The houses data set

We illustrate how PCA can be useful to understand and simplify a data set by using the houses data set. This data set provides census¹ information from all the house block groups from the 1990 California census. A block group has an average of 1425.5 people living in a geographical compact area. This data set also appeared in Kelley Pace & Barry (1997).

The target variable we want to predict is the median house value in a given block. For example, this data set could be used in a project that tries to use the explanatory variables on a block (such as median income, house age, population, etc.) to predict the median house value in that block. To answer this question, we need to apply statistical and machine learning algorithms that are able to understand and model the relationship between the explanatory variables and the target variable (the median house value). Before we can go to this modeling step, we first need to preprocess the data. In this particular example, we investigate if it is possible to reduce the number of explanatory variables. Since in a later stage, we will model the median house value and therefore, this variable will not be part of the PCA. The variables contained in the data set are listed in Table 2.1. A summary of the different variables is given in Table 2.2. In this table, we list the mean, median, minimum value and maximum value of each variable. We also report the 25% quantile (or first quartile) and the 75% quantile (or third quartile). Recall that we assumed that each of the variables has a unit variance. Therefore, we scale the data such that the mean of each of the variables is 0 and the variance is 1.

Before we formally do the PCA, we first investigate the pairwise dependencies between the variables in our data set. One of the objectives of a PCA is to reduce the dimensions of the data set without losing ‘too much’ information. Such a dimension reduction could be useful when two, or more, variables in the original data set have overlapping infor-

¹The Census Bureau’s mission is to serve as the nation’s leading provider of quality data about its people and economy. See <https://www.census.gov>

Variable	Description
mvalue	The median house value.
mincome	The median income.
mage	The housing median age.
rooms	Number of rooms.
bedrms	Number of bedrooms.
popn	Population.
househld	Number of households.
lat	Latitude.
long	Longitude.

Table 2.1: Description of the variables in the houses data set.

Variable	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
mvalue	15000	120000	180000	206890	265000	500000
mincome	0.500	2.560	3.530	3.871	4.740	15.000
mage	1.00	18.00	29.00	28.64	37.00	52.00
rooms	2	1,450	2,130	2,636	3,150	39300
bedrms	1.0	295.5	435.0	538.0	647.0	6450.0
popn	3	787	1170	1426	1730	35700
househld	1.0	280.0	409.0	499.6	605.0	6080.0
lat	32.50	33.90	34.30	35.63	37.70	42.00
long	-124	-122	-118	-119.5	-118	-114

Table 2.2: Summary statistics of the variables in the houses data set.

mation. The correlations between the different variables are shown in Table 3.3. For example, the correlation between the number of bedrooms and the median income is nearly zero. This might be an indication that information about the number of bedrooms does not provide much additional information about the median income in a particular block. This contrasts, for example, with the variables population and household. Indeed, there is a strong correlation of 0.9 between these two variables. This means that knowing the population of a block provides also information about the number of households. Indeed, a large population is a sign for a large number of households in that block.

Since there seems to be a big overlap between the information content of these two variables, using both variables might not be optimal, since the extra space you need to store both variables, will not lead to a sufficient increase in information about each block.

	mincome	mage	rooms	bedrms	popn	househld	lat	long
mincome	1.000	-0.119	0.198	-0.008	0.005	0.013	-0.080	-0.027
mage	-0.119	1.000	-0.361	-0.320	-0.296	-0.303	0.012	-0.074
rooms	0.198	-0.361	1.000	0.930	0.857	0.918	-0.036	0.034
bedrms	-0.008	-0.320	0.930	1.000	0.878	0.980	-0.066	0.066
popn	0.005	-0.296	0.857	0.878	1.000	0.907	-0.109	0.094
househld	0.013	-0.303	0.918	0.980	0.907	1.000	-0.071	0.053
lat	-0.080	0.012	-0.036	-0.066	-0.109	-0.071	1.000	-0.921
long	-0.027	-0.074	0.034	0.066	0.094	0.053	-0.921	1.000

Table 2.3: Correlation matrix of the variables in the houses data set.

In the R code 2.4.1, we show how to perform a PCA on the houses data set. We start with importing the data set and scaling the data. The function `prcomp` then gives the rotation matrix, which we denote here by A . The columns PC1, PC2, etc of the matrix A correspond with the vectors $\underline{\alpha}_1, \underline{\alpha}_2, \dots$ which we derived in Theorem 2.3.3. Once we have the principal components $\underline{\alpha}_k$, we can transform the data. Indeed, in the Examples 2.1 and 2.2, we showed that a data set can be simplified when it is expressed in a new coordinate system. The vectors $\underline{\alpha}_1, \underline{\alpha}_2, \dots, \underline{\alpha}_8$ will serve as the axes in the new coordinate system. The matrix Z then contains the data set, but then expressed in the new coordinate system. If the original data is stored in the rows of the matrix X , then the data set expressed in the new coordinate system is stored in Z and

$$Z = XA.$$

For example, the first observation in the data set is shown in Table 2.4 in the second column. We denote this observation by $\underline{x}_1$. The coordinates of this observation in the new coordinate system are stored in the vector $\underline{z}_1$, which is shown in the third column of Table 2.4. Note that $\underline{z}_1$ can be

calculated as follows:

$$z_{k,1} = \sum_{i=1}^8 \alpha_{k,i} x_i, \text{ for } k = 1, \dots, 8. \quad (2.20)$$

To illustrate why one would be interested in going through this PCA and transforming from the original data set to the new transformed data set, we show in Table 2.5 the variance-covariance matrix of the transformed data set. There are two important observations. Firstly, all covariances are zero, indicating that we transformed the data to uncorrelated variables. Note that the original data was correlated, leading to overlapping information; see Table 3.3. A second important observation is that the diagonal is having values different from one. Indeed, the first variable Z_1 has the largest variance, followed by the second variable Z_2 , etc. Again, this contrasts with the variance-covariance matrix shown in Table 3.3, where all variables had equal variance.

R Code 2.4.1: PCA and the houses data set

```

Data= read.csv("houses.csv", header=TRUE)
names(Data)=c("mvalue", "mincome", "mage",
              "rooms", "bedrms", "popn", "househld", "lat", "long")

# Scale the data so it has mean zero and variance 1
Data = scale(Data[, -1])

# Apply PCA to the data set
Data.PCA = prcomp(Data)
PC=Data.PCA$rotation
round(PC,digits=3)

      PC1    PC2    PC3    PC4    PC5    PC6    PC7    PC8
mincome -0.045  0.025 -0.891  0.408 -0.053 -0.038  0.178 -0.042
mage     0.218  0.003  0.395  0.889  0.033  0.074  0.016 -0.006
rooms    -0.484 -0.072 -0.091  0.117  0.318  0.507 -0.597  0.156
bedrms   -0.491 -0.053  0.118  0.064  0.376 -0.221  0.229 -0.704
popn     -0.472 -0.020  0.116  0.082 -0.846  0.148  0.024 -0.133
househld -0.492 -0.056  0.110  0.097  0.137 -0.383  0.332  0.676
lat       0.071 -0.703  0.013 -0.073  0.069  0.504  0.485  0.036
long     -0.069  0.702  0.054 -0.065  0.118  0.514  0.462  0.050

Z=Data.PCA$x
round(head(Z), digits=3)

      PC1    PC2    PC3    PC4    PC5    PC6    PC7    PC8
[1,] -2.964 -1.897 -2.111  0.974  0.444 -0.018  0.009  0.262
[2,]  1.952 -1.424 -1.148  2.117 -0.010  0.076  0.116 -0.073
[3,]  1.902 -1.451 -0.349  1.780  0.015  0.004  0.081 -0.059

```

Feature	Original observation $\underline{x}_1$	PCA Transformed Value $\underline{z}_1$
mincome	8.30	-2.964
mage	21	-1.897
rooms	7100	-2.111
bedrms	1110	0.974
popn	2400	0.444
househld	1140	-0.018
lat	37.9	0.009
long	-122	0.262

Table 2.4: Comparison of the original data and PCA-transformed values for the first observation.

The first principal component $\underline{\alpha}_1$ and the second principal compo-

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
PC1	3.90	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PC2	0.00	1.91	0.00	0.00	0.00	0.00	0.00	0.00
PC3	0.00	0.00	1.08	0.00	0.00	0.00	0.00	0.00
PC4	0.00	0.00	0.00	0.82	0.00	0.00	0.00	0.00
PC5	0.00	0.00	0.00	0.00	0.15	0.00	0.00	0.00
PC6	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00
PC7	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.00
PC8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02

Table 2.5: PCA covariance matrix, rounded to two decimal places.

nent $\underline{\alpha}_2$ can be found in the output of R Box 2.4.1, we have that

$$\underline{\alpha}_1 = \begin{pmatrix} -0.045 \\ 0.218 \\ -0.484 \\ -0.491 \\ -0.472 \\ -0.492 \\ 0.071 \\ -0.069 \end{pmatrix} \quad \text{and} \quad \underline{\alpha}_2 = \begin{pmatrix} 0.025 \\ 0.003 \\ -0.072 \\ -0.053 \\ -0.020 \\ -0.056 \\ -0.703 \\ 0.702 \end{pmatrix} \quad (2.21)$$

Taking into account Expression (2.20), one can see that the magnitude of the components of the vector $\underline{\alpha}_1$ determine the impact of the original variables on the new variable Z_1 . For example, for the first observation, we have that the Z_1 value is equal to -2.9 (see Table 2.4). This value is the result of multiplying each original variable with the values in the vector $\underline{\alpha}_1$. For example, we have that $\alpha_{1,1} = -0.045$ which will be multiplied with 8.3, the value for the variable `mincome` of the first observation. Since $\alpha_{1,1}$ is close to zero, the variable `mincome` will have a limited impact on the new variable Z_1 . If we have a further look at the component of $\underline{\alpha}_1$, we see that the $\alpha_{1,3}$, $\alpha_{1,4}$, $\alpha_{1,5}$ and $\alpha_{1,6}$ are relatively large in absolute value, implying that these variables have a substantial impact on the realization of the variable Z_1 . As an approximation for Z_1 we could use (2.20) and write

$$Z_1 \approx \alpha_{1,3}X_3 + \alpha_{1,4}X_4 + \alpha_{1,5}X_5 + \alpha_{1,6}X_6.$$

The corresponding variables are `rooms` (X_3), `bedrms` (X_4), `popn` (X_5) and `househld` (X_6). These variables all give information about the size of the block. Otherwise stated, the Z_1 variable of an observation is mainly determined by the size of the block. Because of the minus sign, large blocks take small (possibly negative) values for Z_1 and the larger the block, the larger Z_1 will be. For the variable Z_2 , we can do something similar based on the values of $\underline{\alpha}_2$:

$$Z_2 \approx \alpha_{2,7}X_7 + \alpha_{2,8}X_8.$$

The variables X_7 and X_8 correspond with the latitude and the longitude of the block. Otherwise stated, the Z_2 variable gives information about the location of the block.

We would like to understand the effect of one of original variables X_j on the new variables Z_k . As we have seen in the previous paragraph, this will help us to interpret the variables Z_k . Instead of studying the principal components $\underline{\alpha}_k$, we can also use the correlation between the original variables and the new variables. For example if $\text{Cor}[Z_1, X_j]$ is close to ± 1 , Z_1 contains a lot of information about the variable X_j , whereas the sign will determine the direction of the dependence. In Table 2.6 we show the correlations of each of the new variables Z_k coming from the PCA with the original variables. For Z_1 , for example, we see strong negative correlations with the variables `rooms` (X_3), `bedrms` (X_4), `popn` (X_5) and `househld` (X_6). If these variables increase (i.e. we consider larger blocks), the Z_1 variable becomes smaller. Hence, by studying the correlations $\text{Cor}[Z_1, X_j]$ we arrive at the same interpretation of the first principal component. We conclude from the second row of Table 2.6 that the second principal component Z_2 is strongly correlated with the location (`lat` and `long`) of the block.

We tried to provide an intuitive explanation of the new variables Z_k . First, we showed that understanding Z_k could be done by studying the principal component $\underline{\alpha}_k$. Subsequently, we investigated correlations between the variable Z_k and the original variables to arrive at the same

	mincome	mage	rooms	bedrms	popn	househld	lat	long
Z_1	-0.09	0.43	-0.96	-0.97	-0.93	-0.97	0.14	-0.14
Z_2	0.03	0.00	-0.10	-0.07	-0.03	-0.08	-0.97	0.97
Z_3	-0.92	0.41	-0.09	0.12	0.12	0.11	0.01	0.06
Z_4	-0.37	-0.80	-0.11	-0.06	-0.07	-0.09	0.07	0.06
Z_5	0.02	-0.01	-0.12	-0.15	0.33	-0.05	-0.03	-0.05
Z_6	-0.01	0.02	0.15	-0.06	0.04	-0.11	0.14	0.15
Z_7	0.04	0.00	-0.14	0.05	0.01	0.08	0.11	0.11
Z_8	-0.01	0.00	0.02	-0.09	-0.02	0.08	0.00	0.01

Table 2.6: Correlation between the variables Z_k of the principal components and the original variables in the houses data set

conclusions. However, these two approaches can be linked by using the following formula:

$$\text{Cor}[Z_k, X_j] = \sqrt{\lambda_k} \alpha_{k,j}.$$

This formula confirms that studying the correlations between the original variables and the principal components is equivalent with studying the vectors $\underline{\alpha}_k$.

Figure 2.6 shows a scatterplot of the variables Z_1 and Z_2 (in grey). Note that the correlation between these two variables is zero. Note, this does not necessarily mean that the variables are independent. However, from the plot we can conclude that the variables are only weakly dependent. The blue arrows for each variable X_k correspond, up to a scaling factor, with the vector $(\alpha_{k,1}, \alpha_{k,2})$ and visualize the effect of each variable on the variables Z_1 and Z_2 . For example, the variable `long` points in the positive direction of Z_2 , indicating that the values of `long` have a strong and positive effect on the variable Z_2 . The variable `long`, on the other hand, has a small effect on Z_1 . The corresponding arrow is indeed not really pointing in the Z_1 direction.

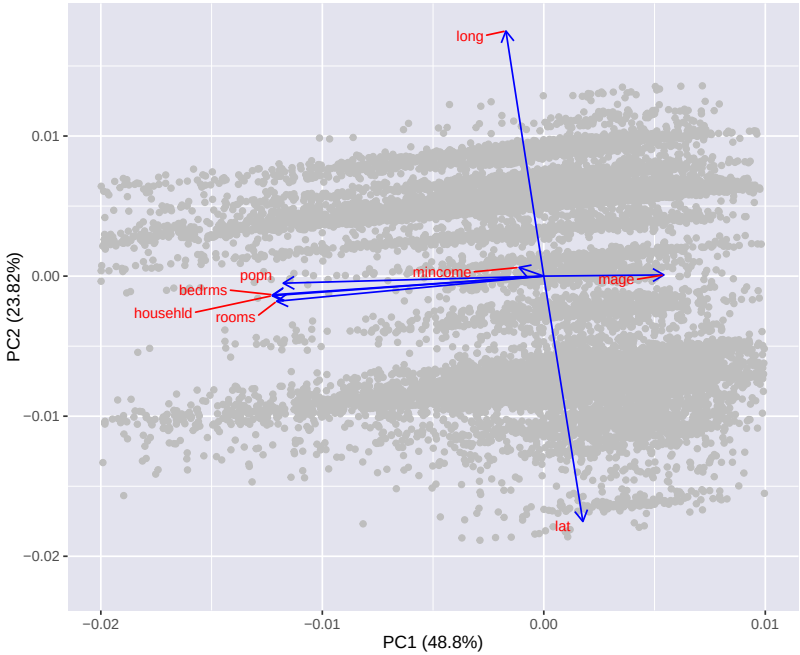


Figure 2.6: Scatterplot with the variables Z_1 and Z_2 together with the effect (blue arrows) of the original variables on the new variables Z_1 and Z_2 .

R Code 2.4.2: PCA and the houses data set

```
summary(Data.PCA)
Importance of components:
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
Standard deviation	1.97	1.38	1.03	0.90	0.380	0.28	0.22	0.120
Proportion of Variance	0.48	0.23	0.13	0.10	0.018	0.01	0.006	0.002
Cumulative Proportion	0.48	0.72	0.86	0.96	0.98	0.99	0.998	1.000

So far, we accomplished one goal: we have transformed the data set such that we can express the data using new variables which are easier (uncorrelated) and such that most of the variation is captured by the first variables. A second goal of the PCA study is to reduce the number of variables. In the R box 2.4.2, we show how you can obtain the standard deviations of each of the variables Z_k . For example, we find that $\text{Var}[Z_1] = 1.97^2$. Taking into account (2.16), we have that the sum of all the variances is equal to 8. The proportion of variance explained corresponds with formula (2.18) and shows how much of the total variance is explained by one of the principal components. For example, using the first principal component we can explain 48% of the total variability. The cumulative proportion of variance explained is determined using Expression (2.19). For example, if we only use Z_1 and Z_2 and remove the remaining six variables, we will lose information about the variability of the data. However, the table shows that we only lose 28%, i.e. the first two component already capture 72% of the variability.

The question is how many principal components (i.e. how many variables Z_k) we should keep. From the table in R box 2.4.2 it is clear that we prefer not to use all eight variables, since the last ones (Z_7 and Z_8) are adding very little to the total variability, so keeping these variables in the data set could in some cases be a 'waste of space'. On the other hand, only keeping one variable might result in a loss of information which is too large. One possibility is to determine first the proportion of total variability you would like to keep after reducing the number of variables in the data set. For example, if one would state that keeping 95% of the variability is sufficient, it follows from R box 2.4.2 that we

need to keep the variables Z_1, Z_2, Z_3 and Z_4 .

Proportion of variance explained for determining the number of PCs.

For a given $\delta \in [0, 1]$, we will keep k variables if k corresponds with the smallest number such that

$$\frac{\sum_{i=1}^k \lambda_k}{\sum_{i=1}^p \text{Var}[X_i]} \geq \delta.$$

An alternative for determining how many variables one should keep after performing a PCA, is to use a screeplot, which plots the variance of each of the variables Z_k on the y -axis and the index k on the x -axis, see Figure 2.7. A screeplot will mostly have a convex shape, since the variance explained of the different principal components declines rapidly in the beginning and slows down near the end. For example, when we go from $k = 1$ to $k = 2$, the variance drops significantly, indicating that Z_1 is an important variable explaining a substantial part of the variance. Also going from $k = 2$ to $k = 3$ indicates that Z_2 is an important variable. However, if we go from $k = 5$ to $k = 6$, and further, we see that the plot is rather flat, indicating that these variables are not very distinctive w.r.t. the variance explained. As a result, the most interesting variables are Z_1, Z_2, Z_3 and Z_4 .

Screeplot Rule for determining the number of PCs.

The last component to keep is the one before the plot begins to straighten out into a horizontal line.

The variances of the variables in the original data set are all equal to 1, and therefore each variable contributes an equal share to the total variability of the data set, which is equal to p ; see Expression (2.15). If we have a look at Figure 2.7 and the table in R box 2.4.2, we observe that each Z_k contributes a different share of the total variability. Moreover, from (2.14), we know that the variances correspond with the eigenval-

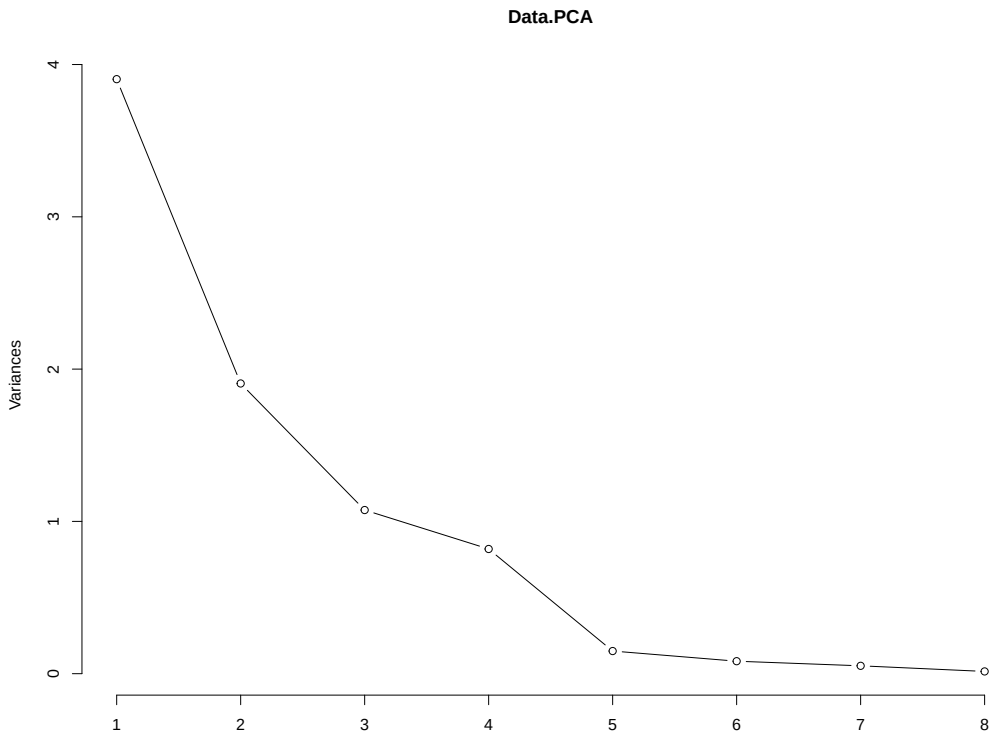


Figure 2.7: Screeplot for the housing data set. Each point indicates the variance of one the variables Z_k .

ues. Some principal components Z_k have variance larger than one, implying that they contribute more than the original variables, but some variables Z_k have variances smaller than one, indicating they are less informative than the original variables when explaining the total variability. The eigenvalue criteria for selecting the number of principal components to keep in the data set then stipulates that you should only keep the variables having a variance larger than one. In our data set, this means we keep the variables Z_1, Z_2 and Z_3 .

Eigenvalue Rule for determining the number of PCs.

The number of variables to keep is equal to k if $\lambda_1, \dots, \lambda_k \geq 1$ and $\lambda_{k+1}, \dots, \lambda_p < 1$.